

CITY UNIVERSITY OF HONG KONG

Application Form for Huawei Seed Grant

1 Name of the applicant, position and home Department, and the date of joining the University

- Name of applicant: Chen MA (00473705)
- Position and affiliated department: Assistant Professor, Department of Computer Science
- Date of joining CityU: August 25, 2021
- Duration of the project: 2 years

2 Title of the Proposed Research

- English Project Title: Towards Diversified Recommender Systems
- Chinese Project Title: 面向多样化的推荐系统

3 Abstract of the Project

Personalized recommender systems are playing an important role in modern society to reduce information/choice overload, satisfy personalized demands of users, and increase the revenue of service/product providers. Despite being helpful, recommender systems are criticized as a primary cause of information “filter bubble”, where users are trapped in an algorithm-created bubble without the opportunity to view diverse contents. This can lead to severely negative consequences like fake news proliferation and political polarization. Although recent studies have managed to improve the recommendation diversity and break filter bubbles, there still lack general solutions to address the accuracy-diversity dilemma.

In this project, we will conduct systematic investigations on diversifying personalized recommendations and breaking filter bubbles in recommender systems. We plan to diversify recommended items against filter bubbles. Improving the recommendation diversity is a silver bullet, however, will induce the accuracy-diversity dilemma. To solve this problem, we will build a model-agnostic multi-objective framework and treat diversity as one of the objectives. Moreover, we will also investigate the exploitation-exploration trade-off of contextual multi-armed bandit to solve the accuracy-diversity dilemma.

4 Key Objectives

Tackle the accuracy-diversity dilemma when breaking filter bubbles by diversification.

- Improving recommendation diversity is a silver bullet to prevent users from trapping in the filter bubble.
- Exploring the exploitation-exploration trade-off to address the accuracy-diversity dilemma.

5 Research Methodology

Background. With the rapid growth of Internet services and mobile devices, personalized recommender systems have emerged and they are beginning to play an important role in modern society. They can: (i) help users easily discover products that they are interested in, and (ii) create opportunities for product providers to increase the revenue. Nowadays, recommender systems are extensively deployed in Internet services, for example, around 80% of watching actions in Netflix and 35% of sales in Amazon are brought by their recommendation engines.

Though helpful and effective, recommender systems are criticized as a major driving factor of creating the *Filter Bubble* [1]: users are trapped inside an information bubble controlled by personalized algorithms or filters. This can create significant information barriers and prevent users from seeing different viewpoints, thus leading to many negative consequences including over-specific information [2], user boredom [3], fake news proliferation [4], and even political polarization [5]. Specifically, the U.S. presidential election result in 2016 is also associated with the filter bubble influence of social media like Twitter and Facebook [6].

To break the filter bubble in recommender systems, improving the recommendation diversity is regarded as one essential means to drag users out of filter bubbles [7, 8]. But diversified recommendations take risk of downgrading the recommendation relevance/accuracy, which will induce the accuracy-diversity dilemma [9, 10]. Thus, how to balance the accuracy-diversity trade-off is important.

To address the aforementioned issue, we define the following task of this project: 1) Design a recommendation diversification framework to **address the accuracy-diversity dilemma**, where any recommendation model and diversity metric can be plugged in; 2) Furthermore, an exploration strategy will be investigated to help users **discover new interests**.

5.1 Research on Diversifying Recommendations against Filter Bubbles

After establishing the understanding of filter bubbles, we will investigate how to improve the recommendation diversity to break the filter bubble. Although many studies [11, 12] focus on the diversity enhancement, they do not comprehensively address the accuracy-diversity dilemma. Moreover, few of them manage to explore new interests of users as a means of diversity boosting. To tackle these challenges, we plan to: 1) design a general diversity enhancing framework that either recommendation model can be plugged in, and 2) design dedicated strategies to help users explore and diversify their interests.

5.1.1 Subtask on Diversifying Recommendations as Multi-Objective Optimization

Our preliminary analysis shows solely optimizing the recommendation accuracy harms the recommendation diversity. The limit comes from the core idea of most recommendation models: recommending similar items to those that users have preferred. Although this is a safe strategy, it squeezes users' viewing scopes and downgrades the recommendation diversity, which leads to a trade-off between accuracy and diversity [9, 10]. To balance this trade-off, we seek to design a model-agnostic framework that a broad community of recommendation models and diversity metrics can be plug-and-play.

Naturally, we treat the recommendation diversity loss $\mathcal{L}_2(\cdot)$ as another important objective along with the accuracy $\mathcal{L}_1(\cdot)$: $\min_{\Theta} \mathcal{L}(\Theta) = [\mathcal{L}_1(\Theta), \mathcal{L}_2(\Theta)]$. A solution is considered as optimal for both objectives in the sense that no objective can be further improved without hurting the other one. This optimality is widely acknowledged in multiple-objective optimization and named as *Pareto efficiency* or *Pareto optimality*. In the context of Pareto efficiency, solution A is considered to dominate solution B only when A outperforms B on all the objectives, i.e., $\mathcal{L}_1(\Theta_A) \leq \mathcal{L}_1(\Theta_B)$ and $\mathcal{L}_2(\Theta_A) \leq \mathcal{L}_2(\Theta_B)$. The aim of Pareto efficiency is to find solutions that are not dominated by any others. One typical approach to solve this problem is *scalarization*, which transforms multiple objectives into a single one with a weighted sum of all the objective functions: $\mathcal{L}(\Theta) = \alpha_1 \mathcal{L}_1(\Theta) + \alpha_2 \mathcal{L}_2(\Theta)$, where $\alpha_1, \alpha_2 > 0$ and $\alpha_1 + \alpha_2 = 1$.

Thanks to recent approaches using Karush-Kuhn-Tucker (KKT) conditions to guide scalarization [13], we plan to build a Pareto-efficient algorithm based on the KKT conditions, which simultaneously optimizes the accuracy and diversity objectives. Motivated by [13], we follow the setting formulate our loss function as: $\min_{\alpha \in [0,1]} \|\alpha \cdot \nabla_{\Theta} \mathcal{L}_1(\Theta) + (1 - \alpha) \cdot \nabla_{\Theta} \mathcal{L}_2(\Theta)\|_2^2$, then there exists an analytical solution for this quadratic problem:

$$\alpha = [(\nabla_{\Theta} \mathcal{L}_2(\Theta) - \nabla_{\Theta} \mathcal{L}_1(\Theta))^{\top} \nabla_{\Theta} \mathcal{L}_2(\Theta) / \|\nabla_{\Theta} \mathcal{L}_1(\Theta) - \nabla_{\Theta} \mathcal{L}_2(\Theta)\|_2^2]_+, \quad (1)$$

where $[z]_+ = \max(\min(z, 1), 0)$ is used to clip the value to $[0, 1]$. Although this is case for two loss functions, this enables the efficient application of the Frank-Wolfe algorithm to generalize to the case of more than two objectives. Assuming that there exist differentiable formulations for each objective correspondingly, we can adopt this scalarization technique to coordinate different objectives into a single objective function and identify a condition for the scalarization weights that ensures the solution is Pareto efficient.

In this subtask, our design will be focused on the following two issues: 1) design differentiable proxies for non-differentiable diversity metrics that can be optimized by the stochastic gradient descent, and 2) design an efficient algorithm that can identify the solution from a set of Pareto optimal solutions.

5.1.2 Subtask on Developing an Exploration Strategy to Discover User Interests

Exploration is an important strategy in recommender systems to recommend items that are out of the scope of users' historical preferences, which can help users explore the undiscovered interests. Different from existing works, we plan to develop an effective exploration strategy from the user aspect, discovering their potential uncover interests against the filter bubble.

We plan to design a diversity-guided contextual multi-armed bandit (CMAB) model to explore users' potential interests. In our setting, each arm $a \in \mathcal{A}$ represents an item featured by a vector $\theta_a \in \mathbb{R}^d$; the context information is represented by $\mathbf{x}_t^u \in \mathbb{R}^d$ including the user u 's preference status and other context information like the spatio-temporal information at round t . Pulling an arm means the associated item is recommended. If the recommended item is tailored to the user u preference (e.g., the recommended item is clicked), then a reward $r_{a,t}^u$ is generated to optimize the arm selection strategy; otherwise, no reward is obtained. The best strategy is to pull the arm with the largest expected reward regarding the context at each trial, and maximize the accumulated rewards in the whole T rounds. This equals to the minimization of

the cumulative regret: $\sum_{t=1}^T \sum_u (\mathbb{E}[r_{a_t^*,t}^u] - \mathbb{E}[r_{a_t,t}^u])$, where $\mathbb{E}[r_{a_t^*,t}^u]$ is the maximum expected reward and a_t^* is the best arm at trial t . One popular reward function assumes a linear mapping function between the expected reward of an arm and the context: $r_{a_t,t}^u = \mathbf{x}_t^{u\top} \theta_a + \epsilon_a$, where ϵ_a is a zero-mean Gaussian noise. To incorporate the diversity guidance, we propose to add one more term to facilitate the diversification exploration: $r_{a_t,t}^u = \mathbf{x}_t^{u\top} \theta_a + \epsilon_a + (1 - \frac{1}{|\mathcal{S}_u|} \sum_{k \in \mathcal{S}_u} \cos(\mathbf{q}_k, \theta_a))$, where the added term measures the average cosine distance between arm (item) a and the whole set of items $\mathcal{S}_u$ that user u has clicked/consumed. The proposed reward function encourages the agent to pull the arm (item) that are both relevant to the user preference but also diversify the user interest. We will apply the Thompson sampling to search for the best pulling strategy, which is a Bayesian approach that dynamically incorporates more information to produce more certain predicted probabilities of each arm with the need to maximize the reward.

We will continue with these tasks: 1) compare the diversity improvement with the proposed multi-objective method (Section 5.1.1), and 2) investigate different priors of Thompson sampling to verify whether a better regret bound and empirical performance can be achieved.

6 Requested Funding and Budget

The total funding requested is \$ 250,000 HKD, which is broken down as follows:

Budget item	Detail breakdown and justification	Estimate cost
Staffing	Recruiting one PhD student Justification: $17,510 \text{ HKD} \times 12 \text{ months} \times 1 \text{ persons} = 210,120 \text{ HKD}$	210,120 HKD
Conference	The research outcomes of this project are expected to be published at top-tier conferences, e.g., ACM SIGKDD.	20,000 HKD
Total		230,120 HKD

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